



Emergent Cooperation in MARL Soccer

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4/23/2026

Soccer is Interesting

Making agents play is even more interesting



Two-Player Zero-Sum Games

- Two-player games where the sum of the payoffs is 0
 - When one player wins, the other loses
- Harder than single-agent RL
 - Not stationary
 - Beating current player may not be optimal



RL “Solves” Two-Player Zero-Sum Games

Magnetic Mirror descent is an update rule proven to linearly converge to a regularize equilibrium in one-shot games.

Empirically shown to be effective in RL settings and extensive form settings. Matches or beats SOTA non-RL methods.

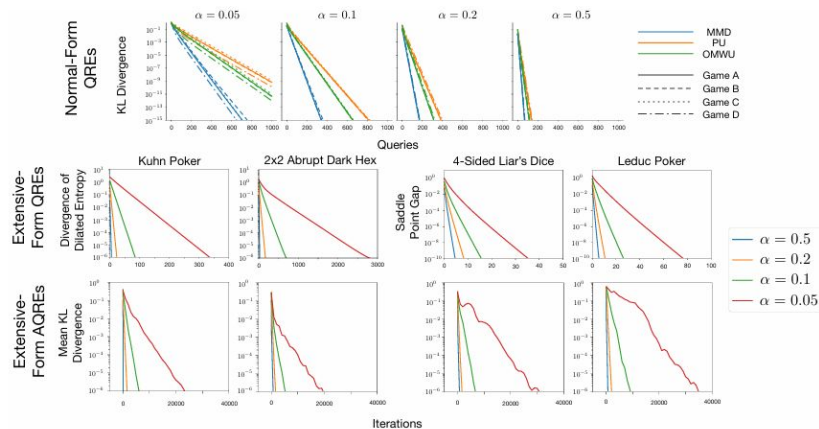


Figure 1: Solving for (A)QREs in various settings.

Teams Are Hard

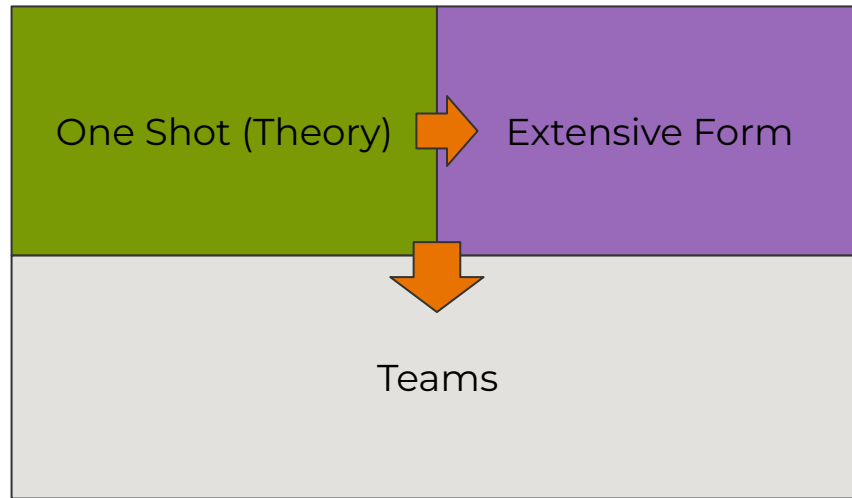
- Soccer is a two player zero sum **TEAM** game
- Members of the team are decentralized
- Example:
 - Suppose we have a players A, A', and B
 - The point of the game is for BOTH A and A' to guess B's number (0,1)
 - A gets to observe B's number A' does not



Teams break minimax

- Minimax Theorem: if player 1 chooses a strategy that maximizes worst-case outcome, the payoff is equal to player 2 choosing a strategy to minimize player 1's best-case outcome
- Teams do not allow arbitrary mixed strategies over joint actions
- Example: modified bit guessing game

Does MMD work with teams?



Regularized RL Algorithm

- Normal PPO Loss + KL to past iterate + KL to entropy
- Anneal regularization and learning rate

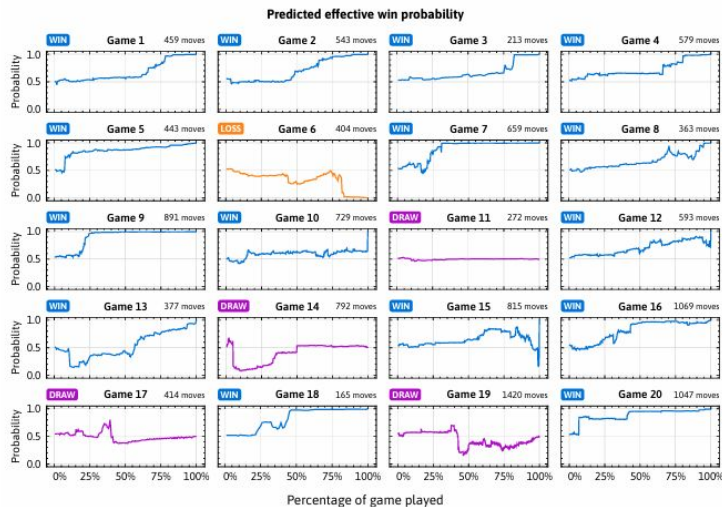
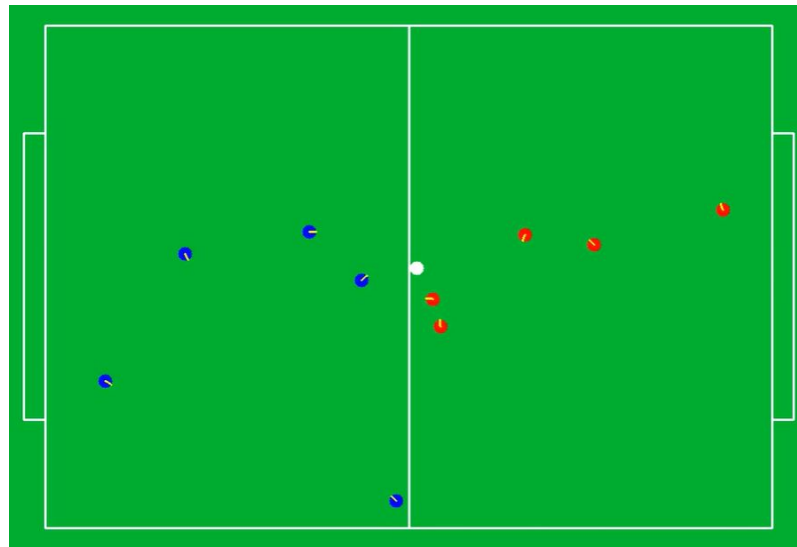


Figure 5: Effective win probability implied by the network value predictions over the games played against Pim Niemeijer. Effective win probability is the expected value of game with winning assigned value 1, drawing assigned value 1/2, and losing assigned value 0.

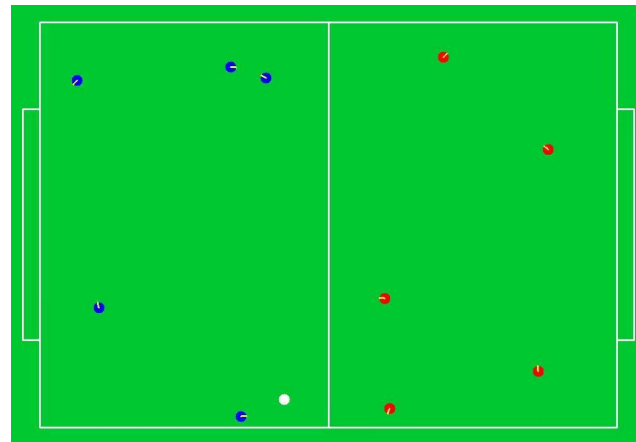
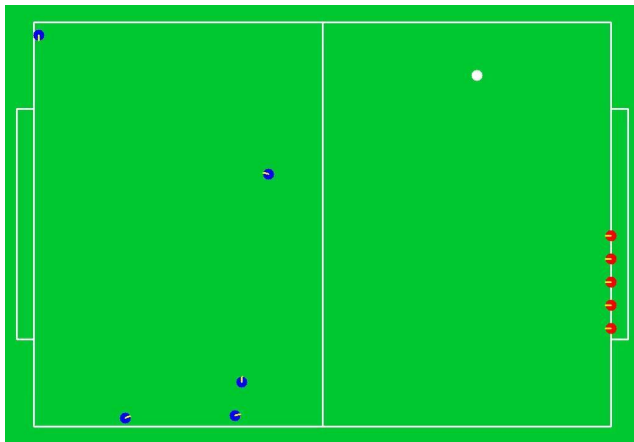
Setting

- Simple fully observable soccer
- Actions: move (forward, back), rotation (left, right), 5 kick strengths
- Tiny MLP: 1 hidden layer



Pure MMD Doesn't Work

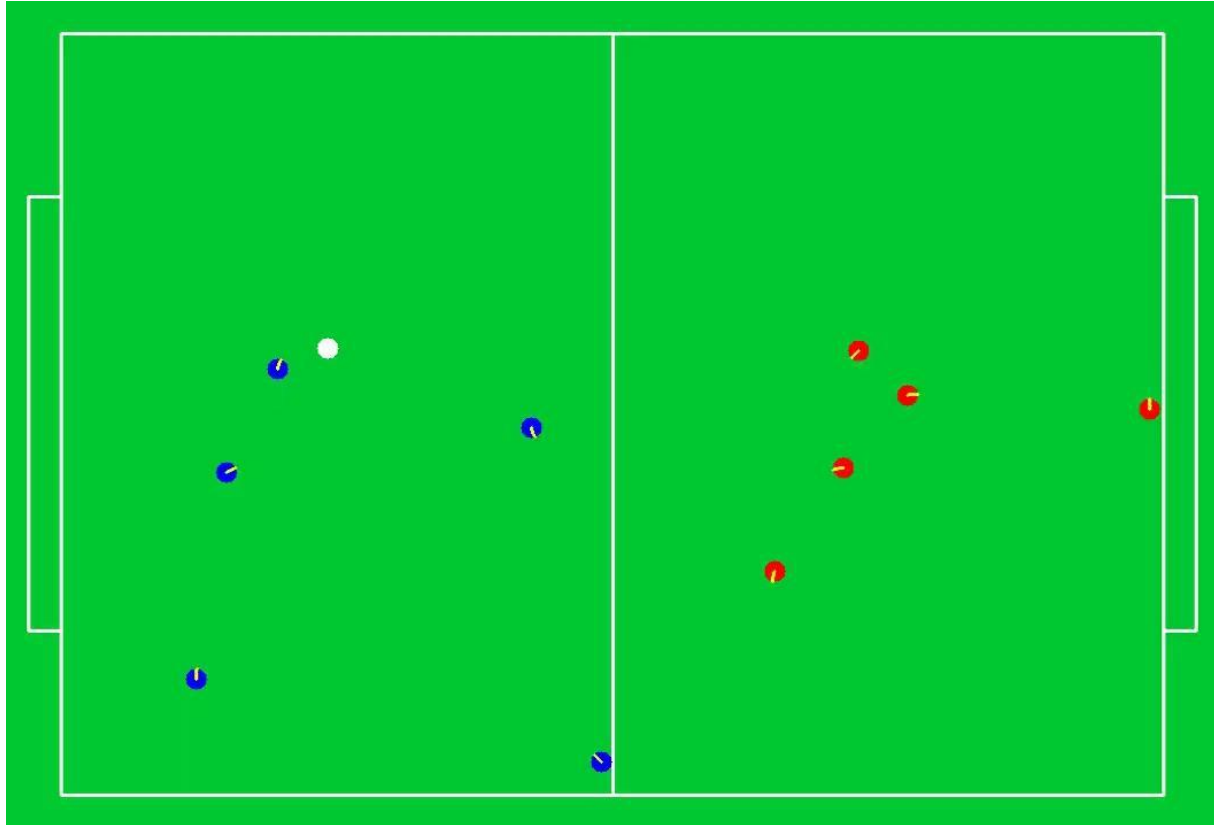
- Rewards are sparse (+1 for score, -1 for scored on)
- Scoring is nontrivial
- Warmstart by training against no opponents with a curriculum of increasing field size



Emergent

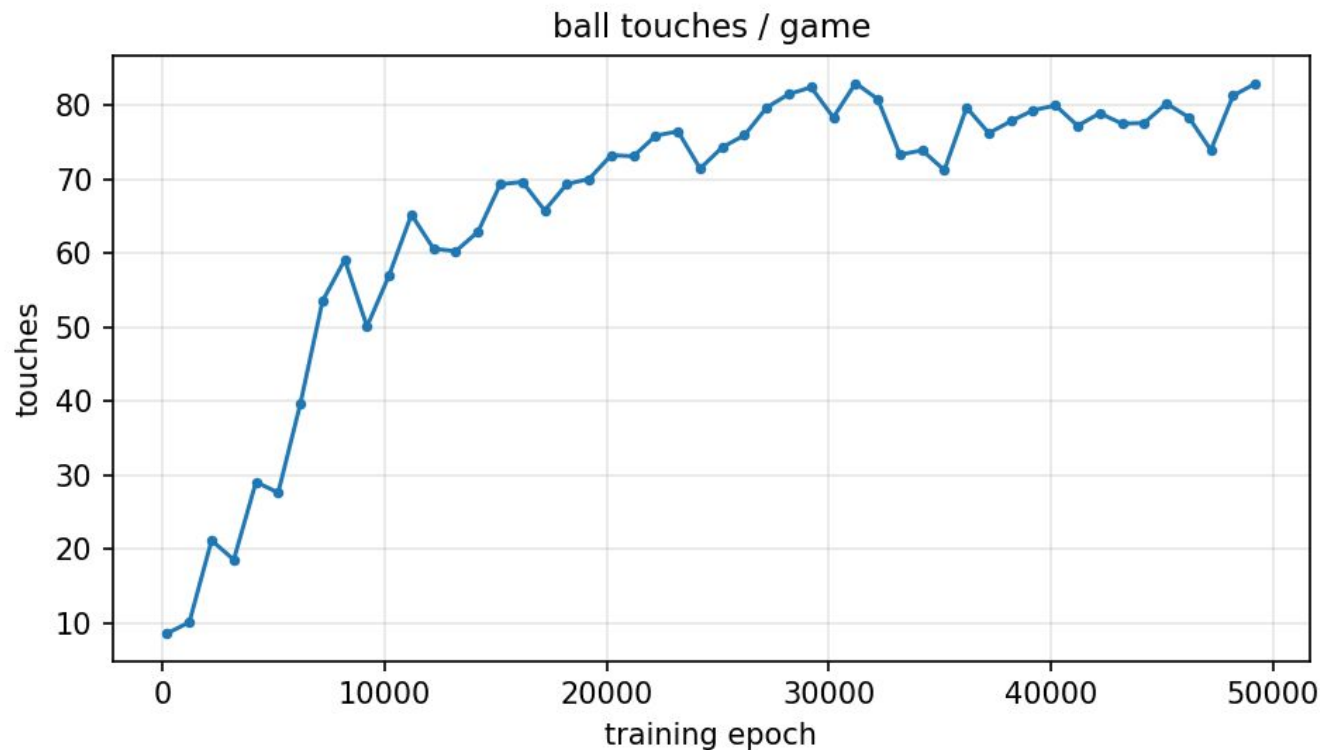
COOPERATION!

Policy Video

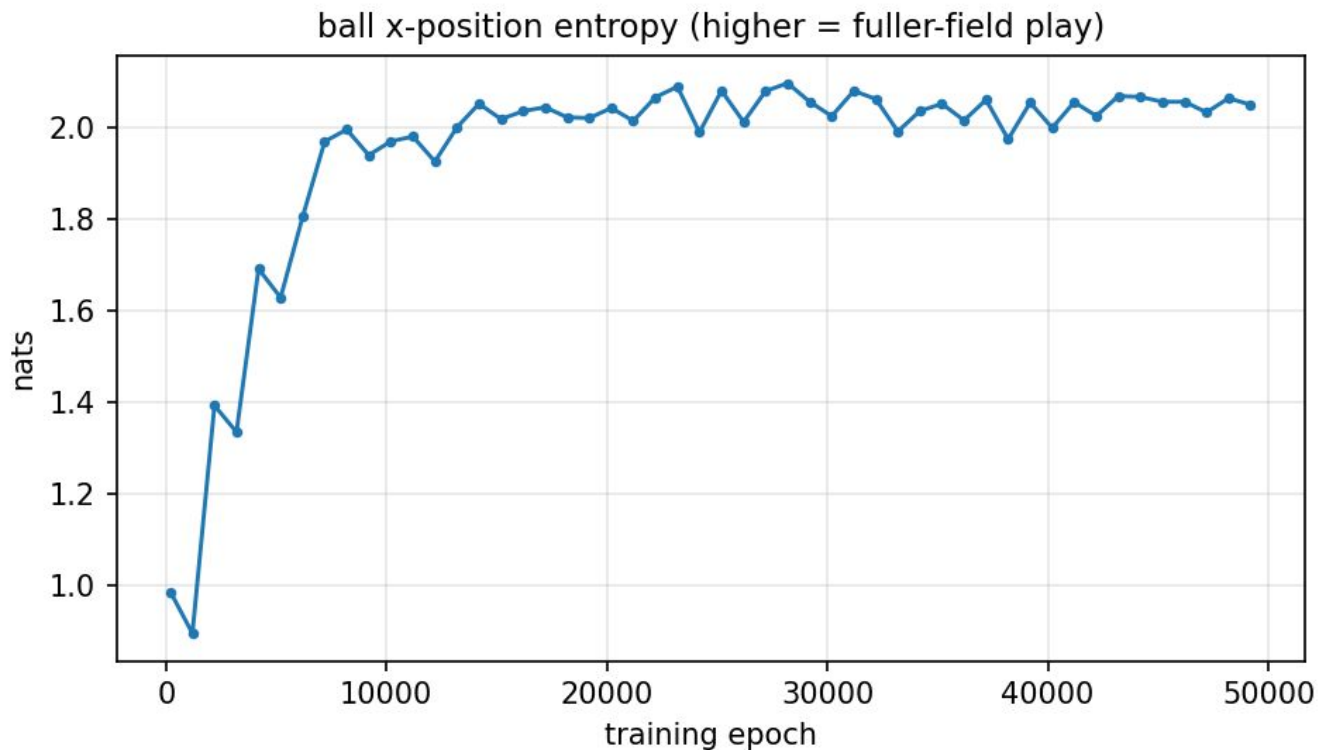


Strategies Learned Over Time

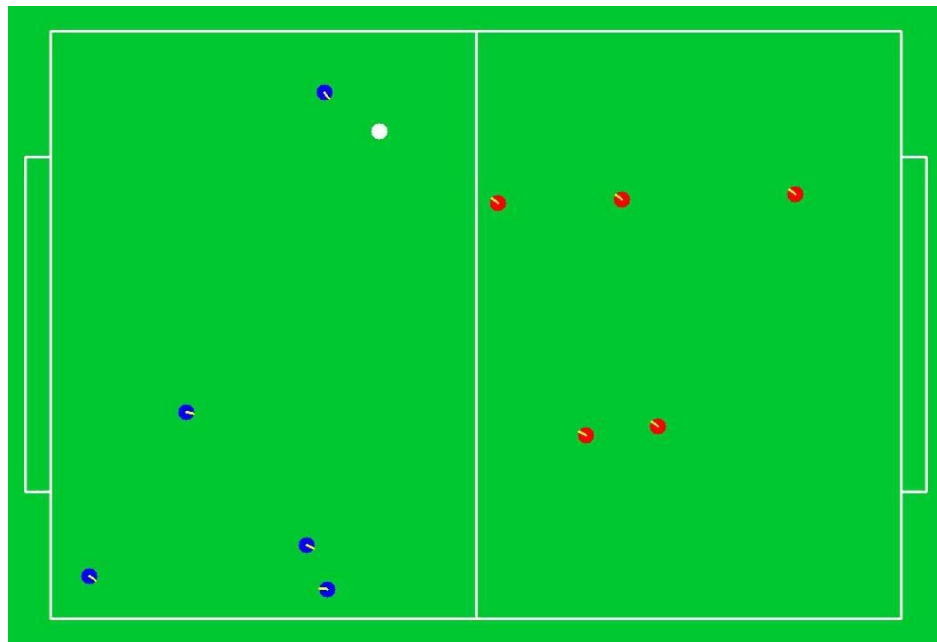
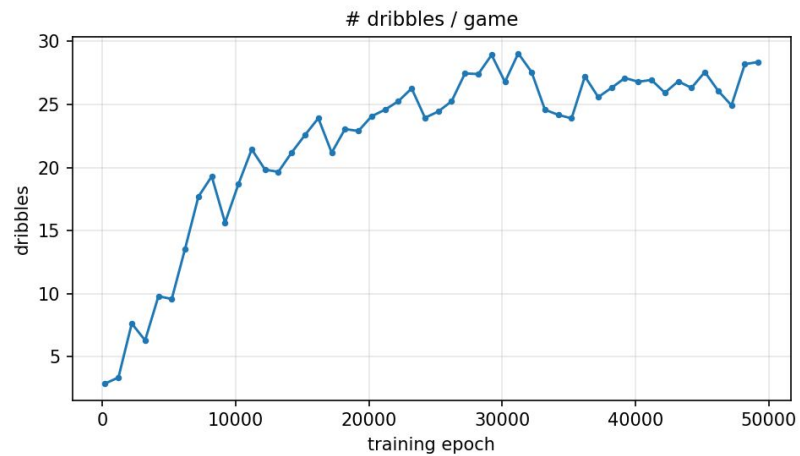
Ball Touches



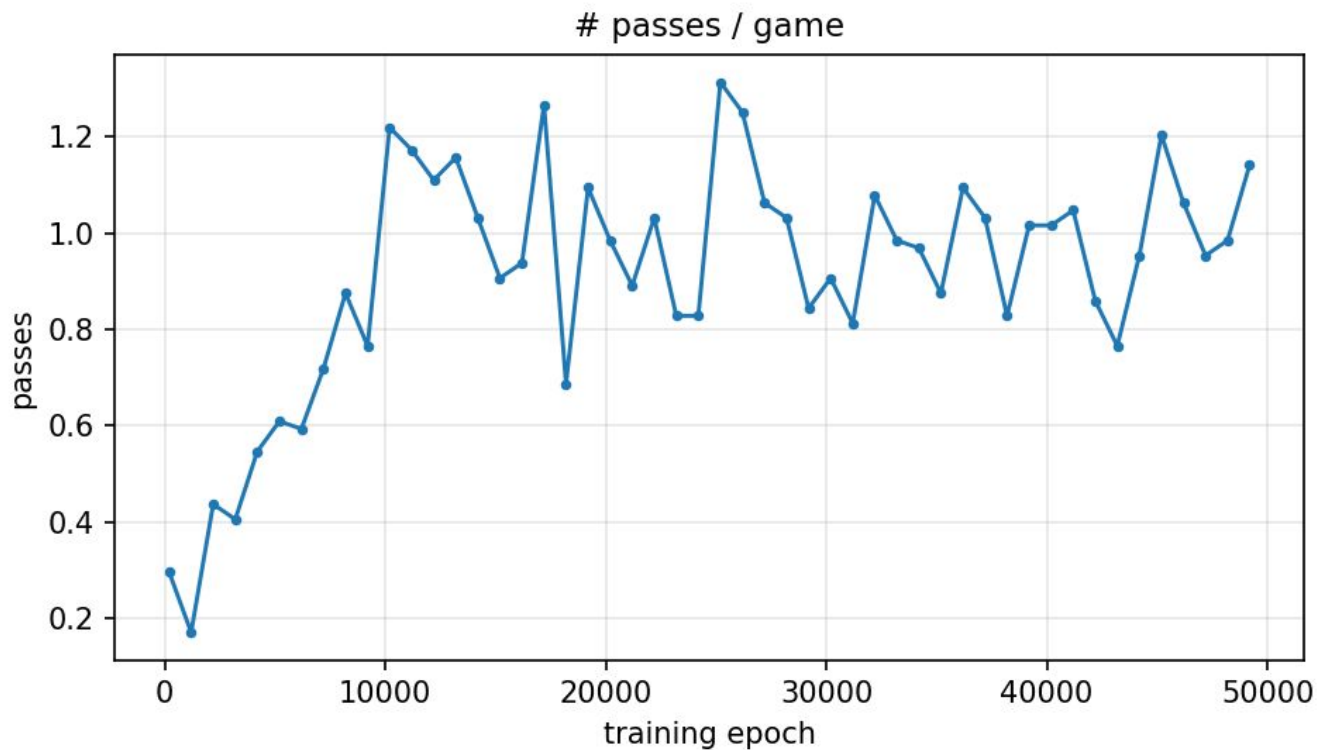
Field Coverage



Dribbling

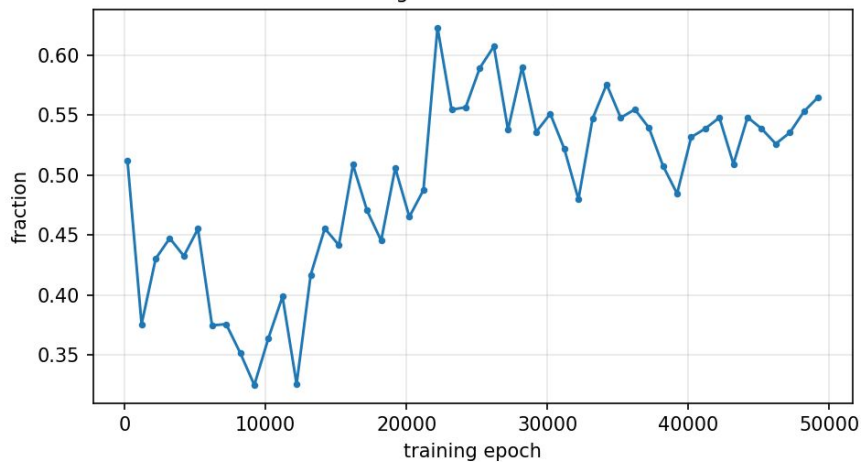


Passing

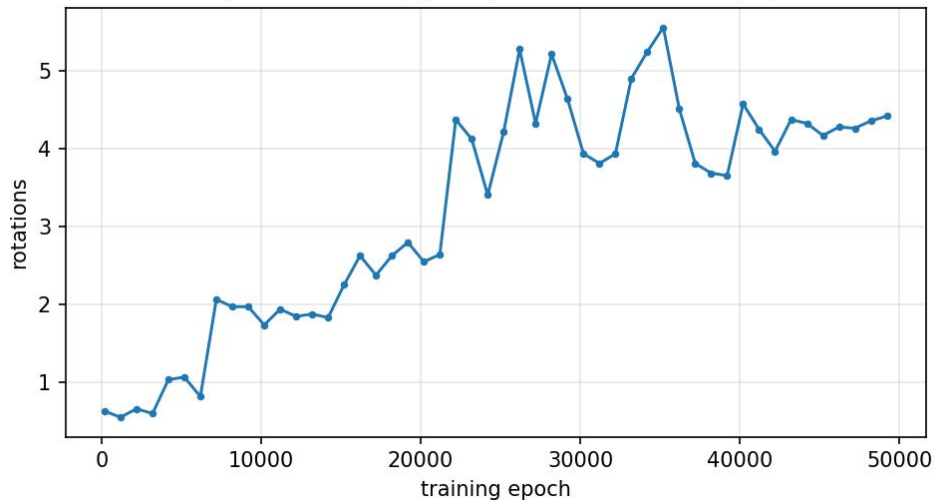


Goalie

% time a goalie is on the baseline

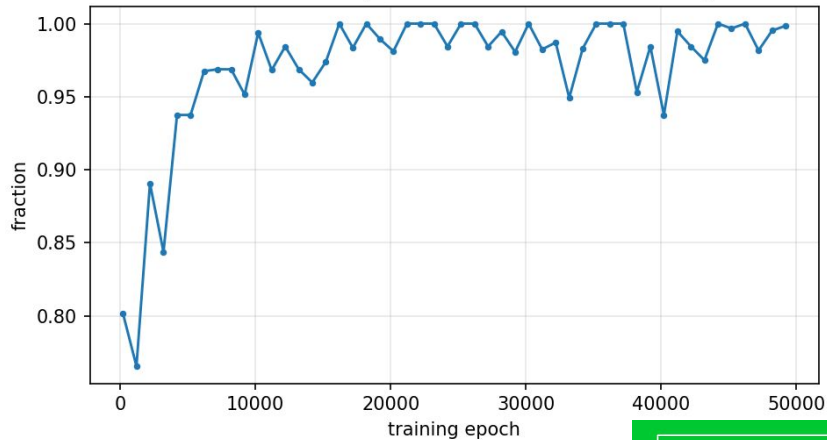


goalie rotations / game (handoff within 20 steps)

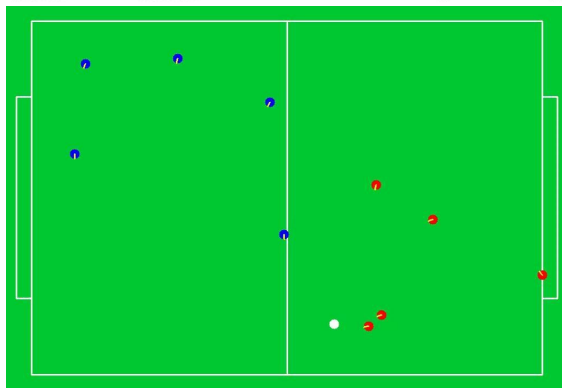
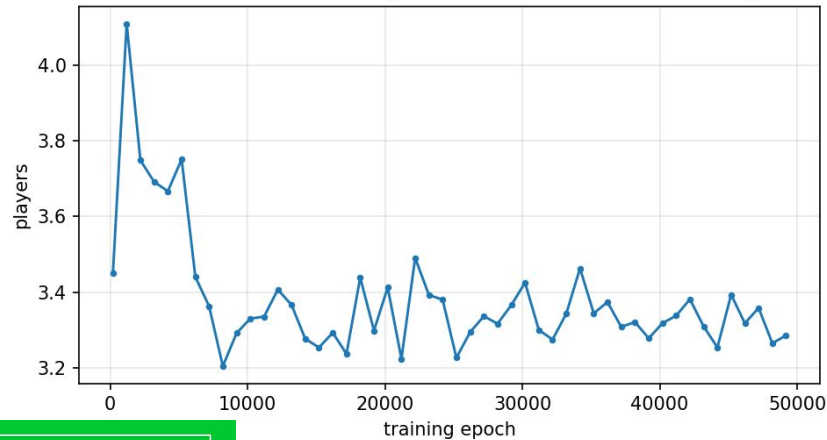


Defensive Players

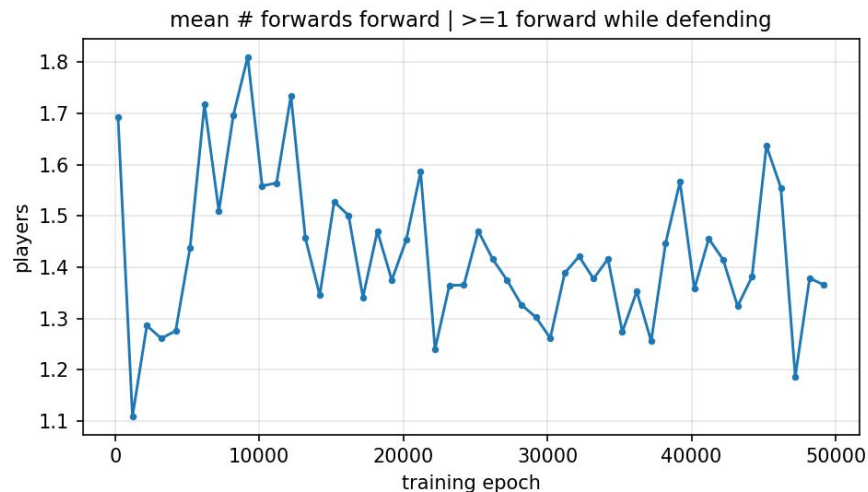
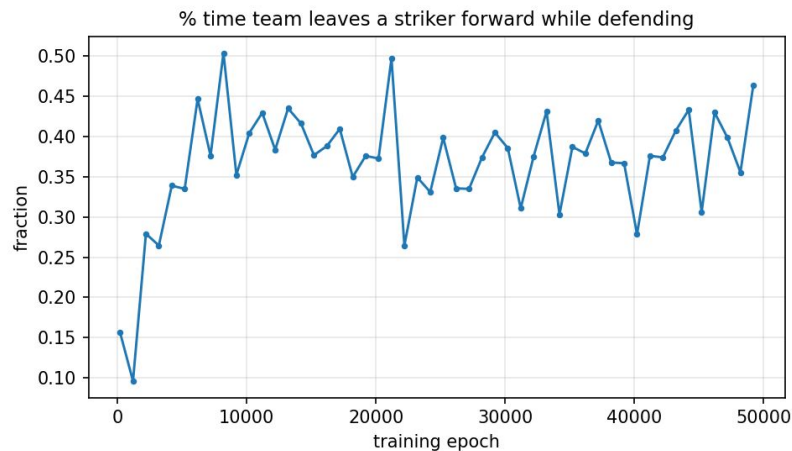
% time team keeps a defender back while attacking



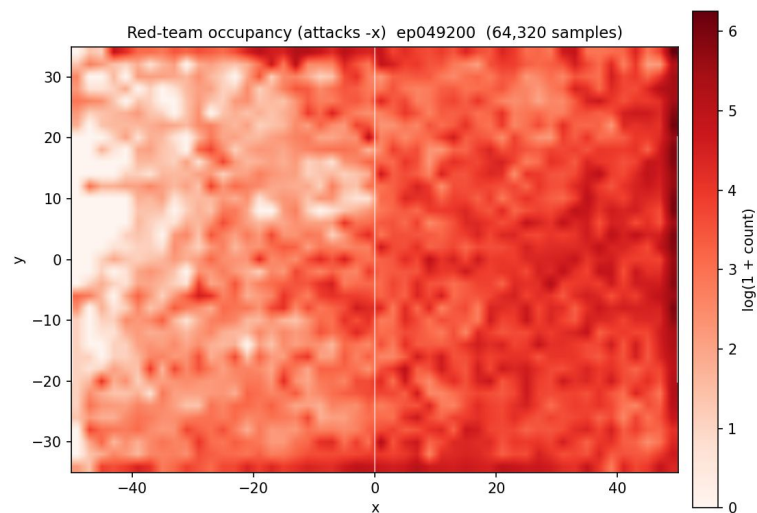
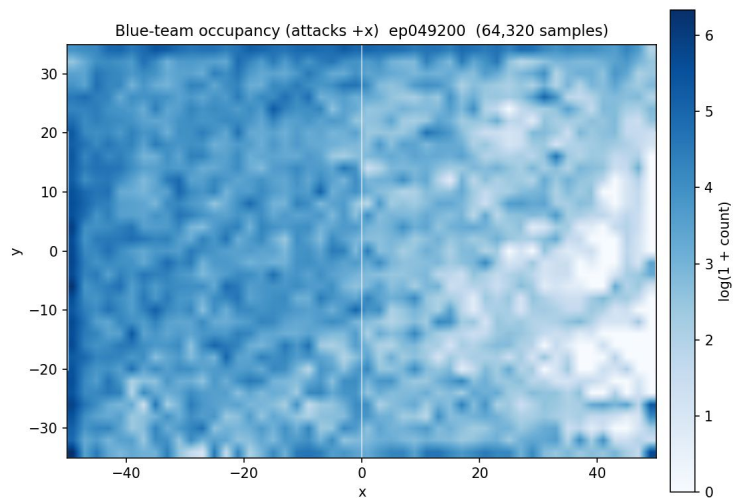
mean # defenders back | ≥ 1 defender back while attacking



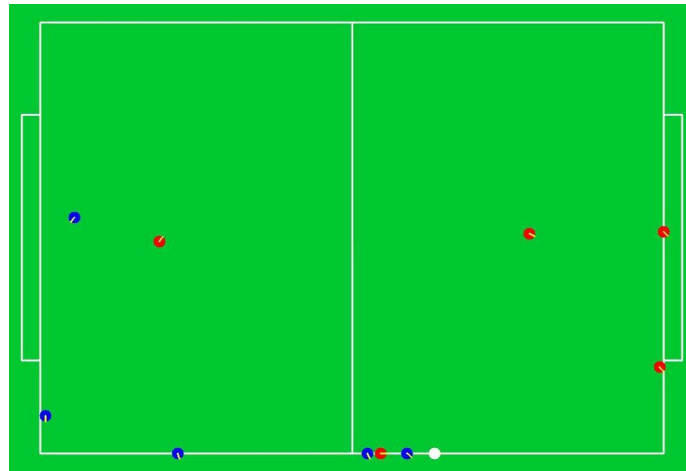
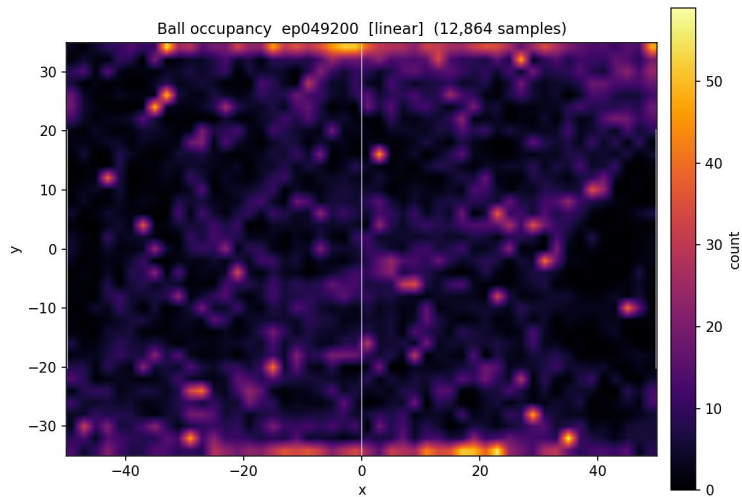
Offesive Players



Bias Towards Defense

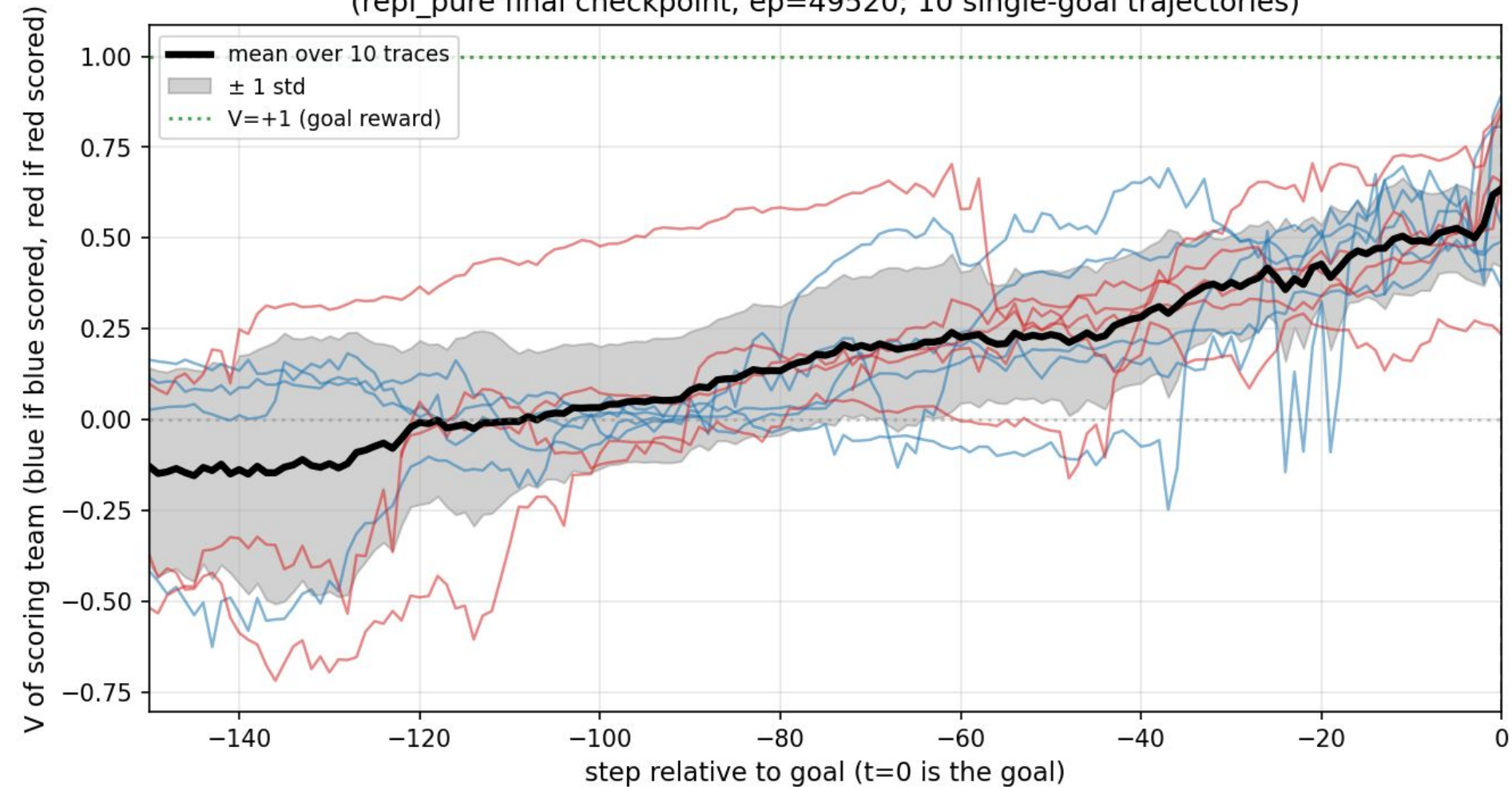


Agents tend to play along the sides

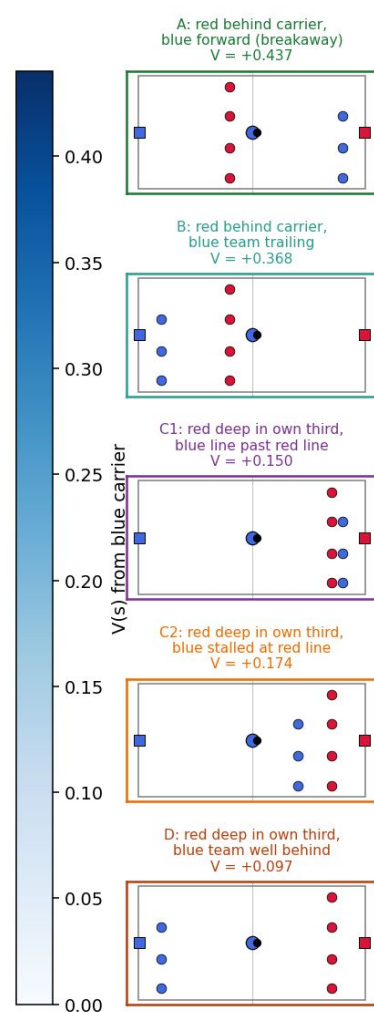
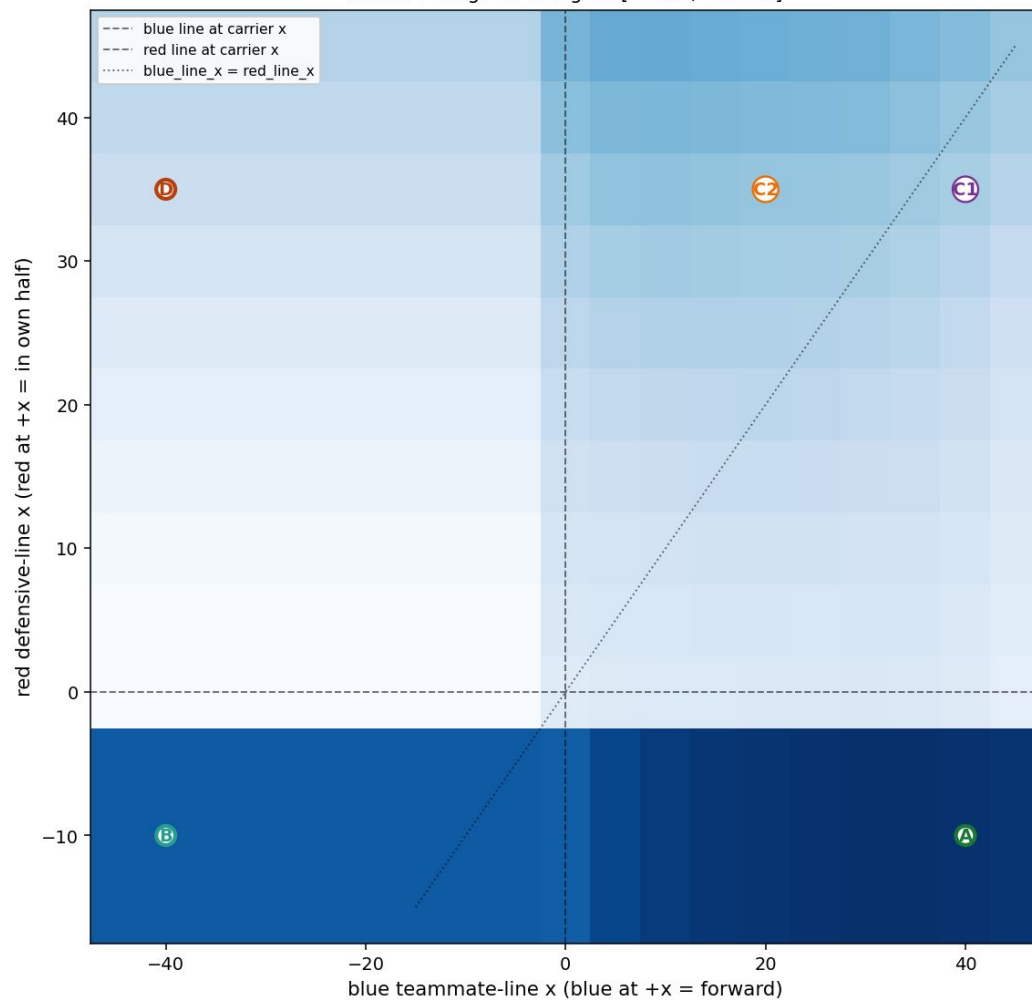


Learned Value

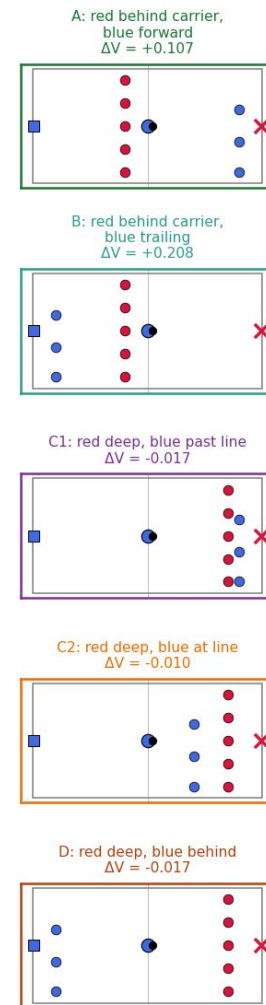
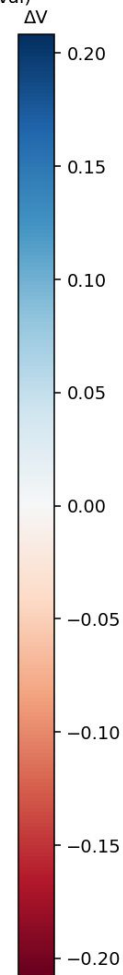
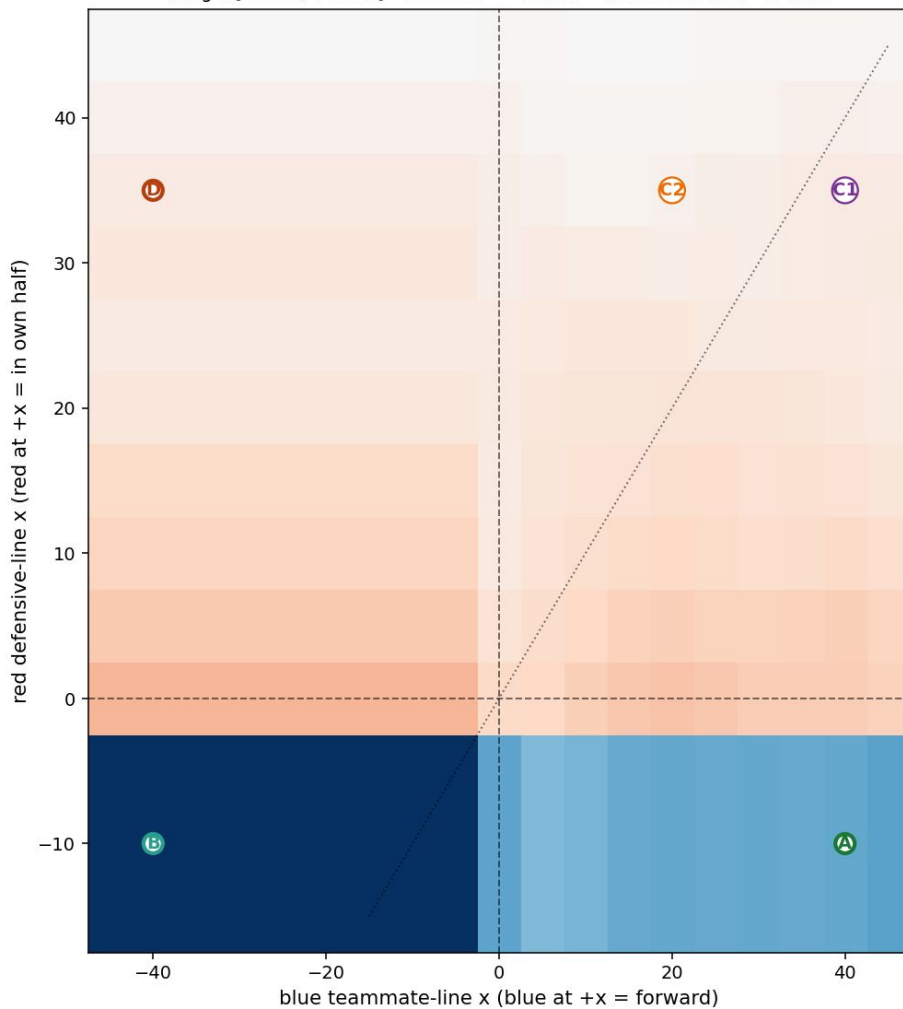
Critic's value estimate rises as the scoring team approaches the goal
(repl_pure final checkpoint, ep=49520; 10 single-goal trajectories)



$V(\text{carrier})$ under varied defensive-line / blue-line positions
(blue carrier at (0,0) with ball; blue+red goalies fixed)
actual V range on this grid: [-0.009, +0.440]

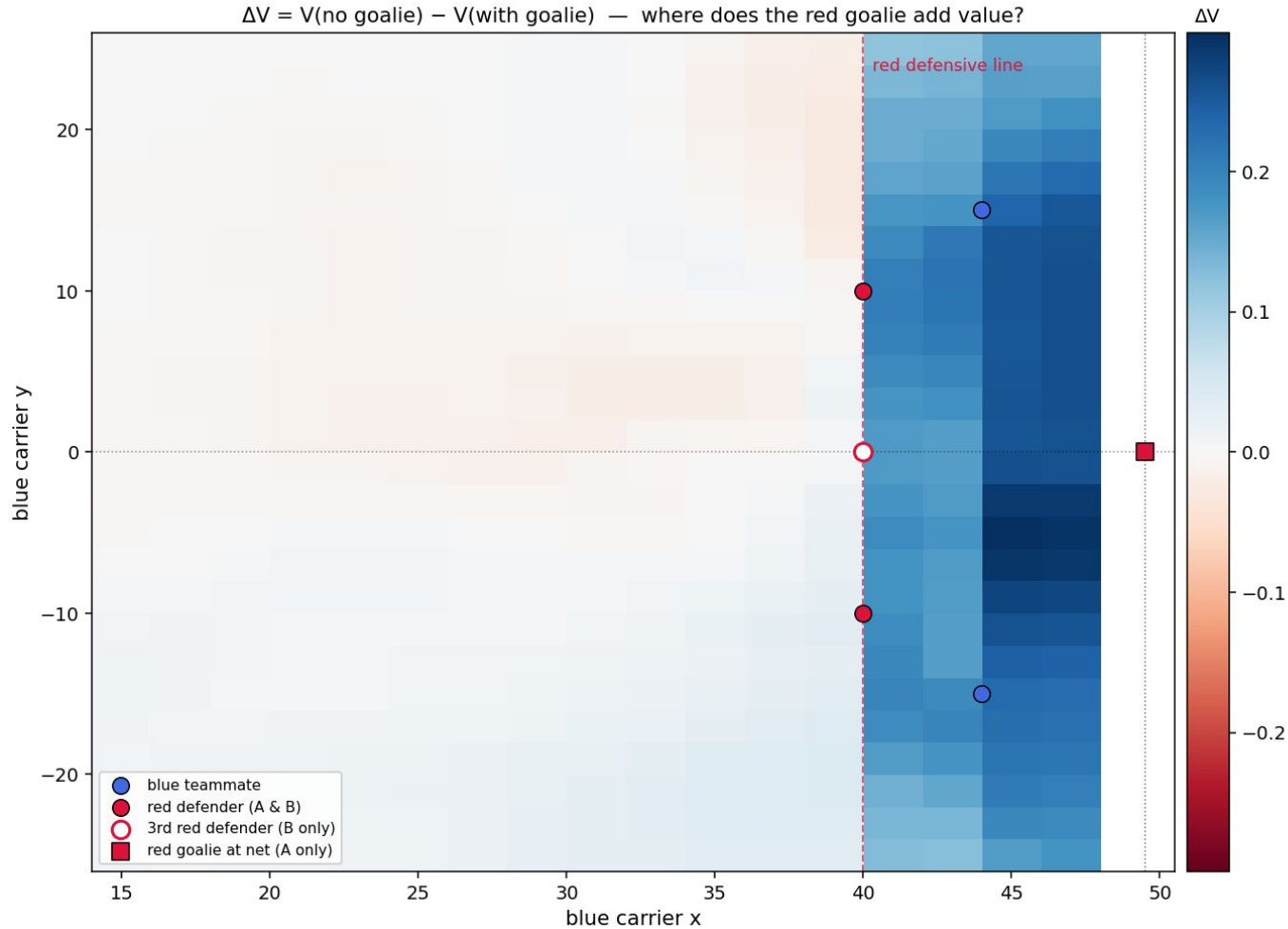


$\Delta V = V(\text{no red goalie}) - V(\text{red goalie in net})$
 blue = goalie is useful for red (blue benefits if removed); red = 5th line defender is better than goalie (red benefits from removal)
 range: [-0.069, +0.208] mean $\Delta V = +0.015$ fraction with $\Delta V > 0$: 23%

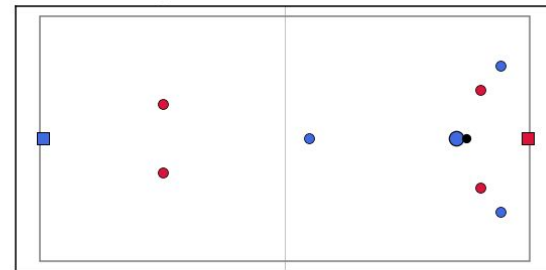


Goalie value in a realistic mid-attack state ($\Delta V \in [-0.027, +0.299]$, mean $+0.050$, 62% of cells positive)

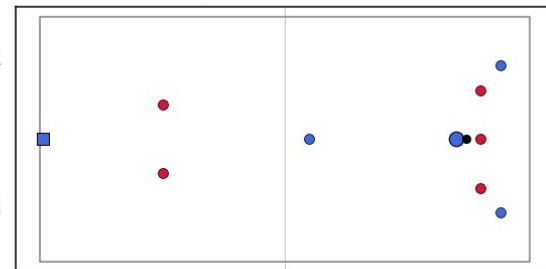
$\Delta V = V(\text{no goalie}) - V(\text{with goalie})$ — where does the red goalie add value?



A: goalie at net (2 line defenders)



B: no goalie (3 line defenders)



Is MMD better than other MARL Algorithms in team settings?

Not sure.

Will wrap up the thesis by comparing against:

- Fictitious play
- League methods

Merely matching these algorithms is already good.



Conclusion

MMD shows emergent cooperation:

1. Passing
2. Playing along the sidelines
3. Goalies and goalie rotations
4. Defensive slant

And it is able to learn good values.

Absolute performance TBD



Thanks

Featuring my goated advisors:
Eugene and Ben

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OUR HISTORY IN TWO PARAGRAPHS



Since its founding in 1831, NYU has been an innovator in higher education, reaching out to an emerging middle class, embracing an urban identity and professional focus, and promoting a global vision that informs its 19 schools and colleges.

Today, that trailblazing spirit makes NYU one of the most prominent and respected research universities in the world, featuring top-ranked academic programs and accepting fewer than one-in-five undergraduates.

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Second Item

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Image Placeholder

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Third Item

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SLIDE WITH IMAGE

Today, that trailblazing spirit makes NYU one of the most prominent and respected research universities in the world, featuring top-ranked academic programs and accepting fewer than one-in-five undergraduates.



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